Agentic AI with LLM-RAG-Based Coordination for Self-Improving Swarm Robotics, SLAM, and IoT

Related Works and Problem Statement

Related Works

I reviewed ten papers that bear on the system I plan to build. I grouped them by the part of the system they inform: LLM-based coordination between robots, collaborative mapping and swarm robotics, learning from experience and retrieval, and small language models. For each paper I note what it does, where it falls short for my purposes, what could come next, and how it connects to my work.

1. LLM-Based Coordination Between Robots

1.1 Building Cooperative Embodied Agents Modularly with Large Language Models (CoELA)

Zhang et al., ICLR 2024. *https://arxiv.org/pdf/2307.02485*

What it does. The authors split each agent into five parts: perception, memory, communication, planning and execution. A language model handles the communication and planning steps, and the memory part stores facts, past events and learned procedures. The agents have no central controller and cooperate by sending each other natural-language messages. In two simulated household-style environments (TDW-MAT and C-WAH), the approach beat planning-based baselines and also worked well alongside human partners. The authors additionally report that an open model they fine-tuned could take the place of the larger one.

Where it falls short. Testing was limited to simulation, with a small number of agents choosing from a set of high-level actions. Memory only helps within a single episode, so the agents do not carry lessons into the next one. Running a large model for every agent is slow and expensive, the model's sense of space is unreliable, and messages are assumed to arrive instantly and without loss.

Next steps. The authors point to finer-grained control and stronger models. My own reading is that larger teams, physical robots and bandwidth-limited communication are the obvious follow-ups.

Why it matters to me. This is the closest existing design to the agent I want to build. I plan to keep its modular structure but make the memory persistent and searchable, so it improves across missions, and to test it under imperfect communication.

1.2 RoCo: Dialectic Multi-Robot Collaboration with Large Language Models

Mandi, Jain and Song, ICRA 2024. https://arxiv.org/pdf/2307.04738

What it does. Every robot has its own language-model agent. The agents talk a task through and settle on a joint plan. Checks from the environment, such as whether an arm can physically reach a spot or whether two arms would collide, are fed back into the conversation before a motion planner carries out the plan. The authors release a benchmark of collaborative manipulation tasks (RoCoBench) in simulation and add a demonstration on real arms.

Where it falls short. The teams are tiny, usually two or three arms. The method leans on the model's grasp of geometry, which is a known weak spot. There is no long-term memory, nothing is learned from earlier tasks, and the cost or failure of communication is not considered.

Next steps. I would expect better spatial reasoning, added memory and learning, and more robots.

Why it matters to me. It shows that robots can reach agreement through dialogue when the plan is checked against physical feasibility. I treat it as the baseline for the reasoning and messaging part of my system, and I intend to add retrieved knowledge and mobile robots.

1.3 SMART-LLM: Smart Multi-Agent Robot Task Planning using Large Language Models

Kannan, Venkatesh and Min, IROS 2024. https://arxiv.org/pdf/2309.10062

What it does. A single instruction is turned into a multi-robot plan in four steps: breaking the task into parts, grouping robots into coalitions, assigning the parts, and executing them. Each step is driven by a few worked examples in the prompt, and the model writes executable code. The authors test on tasks of rising difficulty in the AI2-THOR household simulator and include a real-robot demo.

Where it falls short. Planning is done in one central place. The plan runs mostly without correction, so there is little recovery when something goes wrong, and generated code sometimes contains mistakes. Teams are small and nothing carries over from earlier runs.

Next steps. Plans that adapt during execution, allocation without a central planner, and bigger mixed teams.

Why it matters to me. It offers a clear way to give robots jobs that match their abilities. In my design, the fixed prompt examples could be replaced with allocations retrieved from past missions along with how well they worked out.

1.4 Scalable Multi-Robot Collaboration with Large Language Models: Centralized or Decentralized Systems?

Chen et al., ICRA 2024. https://arxiv.org/pdf/2309.15943

What it does. This paper puts three coordination styles head to head: one central planner, fully decentralized agents, and a hybrid in which a central coordinator is combined with feedback from the individual robots. The authors report that the central planner scales poorly and that the hybrid improves efficiency. Experiments are in simulation.

Where it falls short. The robot counts are small compared with a true swarm, communication is assumed reliable, and there is no lasting memory or learning.

Next steps. Real-robot tests, larger teams and settings where communication is limited.

Why it matters to me. It is the strongest evidence I have for choosing a hybrid architecture, and it gives me a template for comparing coordination styles in my own experiments.

2. Collaborative SLAM and Swarm Robotics

2.1 Swarm-SLAM: Sparse Decentralized Collaborative Simultaneous Localization and Mapping Framework for Multi-Robot Systems

Lajoie and Beltrame, IEEE Robotics and Automation Letters, 2024. https://arxiv.org/pdf/2301.06230

What it does. This is an open-source mapping framework built on ROS 2 for teams of robots with no central server. It works with several sensor types and saves bandwidth by exchanging only the most promising candidate matches between robots rather than everything. It was evaluated on public multi-robot datasets and in real-robot experiments.

Where it falls short. The maps are purely geometric, with no sense of what objects or places are, and nothing reasons over them. Robots still need some connection to each other and enough onboard computing. The rule for choosing what to share is fixed rather than adaptive.

Next steps. Sharing that takes meaning and the current task into account, and communication rules that adapt.

Why it matters to me. This is the SLAM software I expect to build on. It already deals with limited bandwidth, which ties into the IoT and edge side of my title. What I would add is a way to make its maps searchable by a language-model agent.

2.2 Kimera-Multi: Robust, Distributed, Dense Metric-Semantic SLAM for Multi-Robot Systems

Tian et al., IEEE Transactions on Robotics, 2022. (Free copies may be available through Google Scholar.) https://doi.org/10.1109/TRO.2021.3137751

What it does. Robots jointly build a detailed 3D mesh of the environment in which surfaces carry meaning labels. The system chooses reliable matches between robots even when some are wrong, and optimizes the shared map across robots without a central machine. It was tested in simulation and on real multi-robot recordings.

Where it falls short. It needs a lot of computing power and bandwidth. The labels come from a fixed list of classes, the world is assumed not to change, and there is no layer for planning or reasoning.

Next steps. Labels that are not limited to a fixed list, changing environments, and a lighter-weight version.

Why it matters to me. A map with meaning attached is just what a retrieval system needs to index. This paper makes me confident the map can serve as the knowledge source that links my mapping and language-model components.

2.3 Swarm Robotics: Past, Present, and Future

Dorigo, Theraulaz and Trianni, Proceedings of the IEEE, 2021. (May require institutional access.) https://doi.org/10.1109/JPROC.2021.3072740

What it does. A review that explains what makes a robot system a swarm, which collective behaviours have been studied, how swarms are designed, and where they are applied, and it discusses the obstacles to real-world use.

Where it falls short. It was written before language-model agents became prominent, and as a broad overview it goes deep on very little.

Next steps. The paper itself lays out open challenges, which I still need to read closely. I expect that combining learning and high-level reasoning with swarm design is one of them.

Why it matters to me. I use it to ground my swarm terminology, and it supports my point that language-based reasoning and swarm engineering have grown mostly apart.

3. Learning from Experience and Retrieval

3.1 Reflexion: Language Agents with Verbal Reinforcement Learning

Shinn et al., NeurIPS 2023. https://arxiv.org/pdf/2303.11366

What it does. After a failed attempt, the agent writes a short reflection on what went wrong and keeps it in memory. On the next attempt it reads those notes first, so it improves without any retraining. The authors report gains on decision-making, reasoning and coding tasks.

Where it falls short. It only works as well as the model's ability to diagnose its own mistakes, and a wrong reflection can lead later attempts astray. The memory is a small buffer that does not scale, and the experiments involve a single agent in text-based settings.

Next steps. The authors discuss richer memory and combining it with other forms of learning. I see physical and multi-agent settings as the natural extension.

Why it matters to me. It is the simplest way I know to let an agent learn from experience, so I plan to use a version of it as my improvement mechanism, with a proper searchable store in place of the small buffer.

3.2 Retrieval-Augmented Embodied Agents

Zhu et al., CVPR 2024. *https://arxiv.org/pdf/2404.11699*

What it does. The agent looks up relevant past experience from a memory when making decisions, instead of relying only on what is stored in the model itself. I have a high-level understanding of this paper and will confirm its architecture and results when I read it in full.

Where it falls short. To be confirmed. Likely points are a single-agent scope and a memory built in advance.

Next steps. Possibly a memory that keeps growing during operation and is shared between agents.

Why it matters to me. It is the nearest support I have for retrieval-based decision-making in embodied agents. I would extend the idea to a store shared by a team, holding map knowledge and records of past missions.

4. Small Language Models

4.1 Small Language Models are the Future of Agentic AI

Belcak et al. (NVIDIA Research), 2025. Preprint, not peer-reviewed.
https://arxiv.org/pdf/2506.02153

What it does. The authors argue that small models are capable enough, a better fit and cheaper for many of the repetitive, specialized jobs agents do. They suggest systems that use a small model by default and call a large one only when needed, and they describe how to move an existing agent over to small models.

Where it falls short. It is a position paper with limited experimental evidence. It is not about robotics and does not consider communication limits.

Next steps. The authors call for applying and testing the approach more widely. Evaluation on robots is the gap I care about.

Why it matters to me. It gives the reasoning behind using a small model. Robots with limited computing cannot call a large proprietary model at every step, and my project tests whether a small model is good enough in a robotics setting.

	Paper	Venue	Why it fits	PDF / page
1	**Building Cooperative Embodied Agents Modularly with LLMs (CoELA)**, Zhang et al., 2024	ICLR	Closest architecture to yours: memory, communication and planning modules in each agent	https://arxiv.org/pdf/2307.02485
2	**RoCo: Dialectic Multi-Robot Collaboration with LLMs**, Mandi, Jain, Song, 2024	ICRA (IEEE)	LLM-to-LLM negotiation between robots, with real-robot demo	https://arxiv.org/pdf/2307.04738
3	**SMART-LLM: Smart Multi-Agent Robot Task Planning using LLMs**, Kannan et al., 2024	IROS (IEEE)	LLM task decomposition and allocation for heterogeneous robots	https://arxiv.org/pdf/2309.10062
4	**Scalable Multi-Robot Collaboration with LLMs: Centralized or Decentralized Systems?**, Chen et al., 2024	ICRA (IEEE)	Evidence for your centralized vs. decentralized design choice	https://arxiv.org/pdf/2309.15943
5	**Swarm-SLAM: Sparse Decentralized Collaborative SLAM for Multi-Robot Systems**, Lajoie & Beltrame, 2024	IEEE RA-L	Swarm-oriented, bandwidth-aware SLAM; your likely SLAM substrate	https://arxiv.org/pdf/2301.06230
6	**Kimera-Multi: Distributed Dense Metric-Semantic SLAM for Multi-Robot Systems**, Tian et al., 2022	IEEE T-RO	Distributed semantic maps that an LLM or RAG layer could reason over	IEEE page: https://doi.org/10.1109/TRO.2021.3137751 (check for a free author PDF via Google Scholar)

	Paper	Venue	Why it fits	PDF / page
7	**Swarm Robotics: Past, Present, and Future**, Dorigo, Theraulaz, Trianni, 2021	Proceedings of the IEEE	Anchor review for the swarm side	IEEE page: https://doi.org/10.1109/JPROC.2021.3072740 (likely paywalled; try Google Scholar for a PDF)
8	**Reflexion: Language Agents with Verbal Reinforcement Learning**, Shinn et al., 2023	NeurIPS	Cheapest, clearest self-improvement mechanism from past failures	https://arxiv.org/pdf/2303.11366
9	**Retrieval-Augmented Embodied Agents**, Zhu et al., 2024	CVPR	RAG applied directly to embodied agents	https://arxiv.org/pdf/2404.11699 (I'm least sure of this paper's details, so read it before citing)
10	**Small Language Models are the Future of Agentic AI**, Belcak et al., 2025	arXiv preprint, not peer-reviewed	Justification for your custom/small language model	https://arxiv.org/pdf/2506.02153

Problem Statement

Teams of robots are often sent to explore places that have never been mapped, with unreliable wireless links and little computing power on each machine. Collaborative SLAM lets robots build a shared map, and language-model agents let robots reason and divide up work in plain language, but these two lines of research have mostly developed apart. Existing LLM-based multi-robot systems are usually tested in simulation with few robots, assume messages always arrive, depend on large models, and begin every mission from scratch. Collaborative SLAM maps, for their part, were not designed to be searched by a language-based agent.

My research asks whether a small team of robots, each running a lightweight language model, can use a shared and searchable record of the map and of past missions to coordinate better. It also asks whether the team can improve over repeated missions, even when messages are delayed or lost. I will carry out the study in simulation, with 3–5 robots on an exploration or search task.

Research Questions

- **RQ1.** Does retrieval over shared SLAM maps and past experience improve task success and efficiency compared with prompting alone?
- **RQ2.** How does a lightweight, fine-tuned language model compare with a large general-purpose model in task quality, response time and compute cost?
- **RQ3.** Does storing and reusing lessons from failed missions improve performance over repeated missions?
- **RQ4.** How do message delay and message loss affect coordination quality?

Expected Contributions

I expect to contribute an integrated architecture, a searchable representation of collaborative SLAM output, a comparison of small and large language models for team coordination, and an empirical study of experience-based improvement under communication limits.

Objectives

1. Build a simulated multi-robot team of 3–5 robots on a single exploration or search task. Each robot's language-model agent retrieves from a shared knowledge store that holds text descriptions of the map from existing collaborative SLAM software and records of past missions.
2. Fine-tune a small open-source language model (using LoRA or QLoRA) on coordination and planning data generated in simulation, and compare it with a large general-purpose model on task success, response time and compute cost.

3. Add reflection-style memory that stores lessons from failed missions and retrieves them on later attempts, and measure whether performance improves over repeated missions, including under simulated message delay and loss.

Research Type

My work is applied, experimental, design-and-evaluate research carried out through simulation.

- **Applied.** I am not proving a new theory. I am building a working system for a practical problem, coordinating robot teams, out of existing methods: collaborative SLAM, retrieval and small language models.
- **Experimental and quantitative.** I will compare variants and measure the outcomes: with and without retrieval, small versus large model, with and without reflection memory, and different levels of message delay and loss. The results will be numbers such as task success, response time and compute cost.
- **Simulation-based.** All experiments run in a simulator, and real robots are left as optional future work.
- **System integration.** The novelty lies in combining existing components into one framework and testing whether the combination helps, not in inventing a new algorithm. In research-methodology terms, this is close to design science or systems research: I build an artifact and then evaluate it.